Physics in history and society

Textbooks tell the history of physics as a series of brilliant ideas: Galileo, Newton, Maxwell, Einstein. But the ideas did not arise in a vacuum. Someone paid for the instruments, someone did the work without making the front page, wars and industries set the agenda, and some people were kept out. This page looks at physics as a human and social activity – without forgetting that it has nevertheless produced some of the most reliable knowledge we have.

Two ways of looking at the history of science

Historians have long distinguished between an *internal* and an *external* history of science. The internal one follows the ideas: which problems the scientists worked on, which arguments they used, and how one theory replaced another. The external one asks about the surroundings: economics, politics, religion, institutions and technology.

The debate took off in 1931, when the Soviet physicist Boris Hessen gave a sensational paper at the International Congress of the History of Science in London. He argued that Newton's *Principia* was a response to the technical needs of England's emerging capitalism: shipping, ballistics and mining. The French philosopher Alexandre Koyré replied that the Scientific Revolution was above all a revolution in thought – from a closed world to an infinite universe. In 1938 the sociologist Robert K. Merton took a middle position: Puritan values and England's economy created good conditions for natural science, but did not determine the content of its theories.

Today most historians see the two perspectives as complementary. Newton's law of gravitation cannot be explained by England's merchant fleet – but the Royal Greenwich Observatory, which was to supply data for his lunar theory, was built to solve the navy's longitude problem.

Who paid?

Before 1800 there were almost no professional scientists. Science was pursued by university teachers, clergymen, aristocrats with time and money – and by those who could find a princely patron.

The prince’s favour

Tycho Brahe was given the island of Hven by Frederick II and built Europe’s first large research facility – and lost it all when the king died. Galileo named Jupiter’s moons after the Medici family and obtained the post of court mathematician in Florence. The historian Mario Biagioli has shown how the court’s rules of gifts, favour and prestige shaped the very way Galileo presented his discoveries.

The academies

The Royal Society in London (1660) was a private club, financed by members’ subscriptions. The Académie des sciences in Paris (1666), by contrast, was state-funded, and its members were paid by the king to work on tasks the crown needed: maps, canals, clocks and navigation. The two models – free association and state research institution – still exist side by side.

Foundations

Niels Bohr’s institute was built in 1921 with support from, among others, the Carlsberg Foundation, and from 1923 it received large grants from the American Rockefeller Foundation. The foundations made Copenhagen an international centre visited by Heisenberg, Pauli and Dirac – an example of how private foundations can shift the centre of gravity of research.

The state

After 1945 the state became the great paymaster. Vannevar Bush, who had led America’s wartime research, wrote in the report *Science, the Endless Frontier* (1945) that basic research should be funded by the state because it is the source of future technology, health and security. That logic still underlies the research councils.

Scholars and craftsmen

In 1942 the Austrian sociologist Edgar Zilsel argued that modern natural science arose when two groups met: the scholars, who knew Latin and mathematics, and the craftsmen – instrument makers, shipbuilders, compass makers and gun founders – who had practical experience of

materials and measurement. Galileo wrote his *Discorsi* starting from conversations at the Arsenal in Venice.

In the book *Leviathan and the Air-Pump* (1985) Steven Shapin and Simon Schaffer showed that it was not self-evident that an experiment could settle a dispute. When Robert Boyle wanted to prove with his air pump that a vacuum exists, the philosopher Thomas Hobbes objected that the pump leaked, and that an expensive apparatus which only a few could build could not be the basis of certain knowledge. Boyle answered with a new practice: experiments performed before credible witnesses and described in such detail that readers could imagine having been there.

Shapin has also pointed to "the invisible technician". Boyle's air pump was built and operated by Robert Hooke and other assistants, but in Boyle's accounts it is Boyle who does the experiments. The same pattern recurs throughout the history of physics: glassblowers, instrument makers, programmers and technicians, without whose work the famous results would never have come.

Industry: when technology comes before theory

It has often been said that science owes more to the steam engine than the steam engine owes to science. Newcomen and Watt built their engines from experience. Only in 1824 did Sadi Carnot ask how much work can in principle be got out of heat – and his answer became the beginning of thermodynamics (see The Carnot cycle).

The same happened with electricity. The telegraph cables under the Atlantic required precise measurements of resistance and capacitance, and in the 1860s the British Association set up a committee including Maxwell and William Thomson (Lord Kelvin), which fixed the unit of the ohm. Modern SI units have their roots in industry's need to trade in electricity.

The most surprising example is quantum theory. In 1887 Werner von Siemens and Hermann von Helmholtz founded the Physikalisch-Technische Reichsanstalt in Berlin, partly to help the German lighting industry compare light sources. Here physicists measured the radiation from a black body with great precision. It was Heinrich Rubens' and Ferdinand Kurlbaum's new measurements in October 1900 that led Max Planck to find his radiation law – and two months later to introduce the energy quantum.

War and big science

War has always been a driving force for technical science: ballistics, navigation, metallurgy. But the 20th century changed the relationship between physics and the state for ever.

The First World War

Often called the chemists' war. Fritz Haber, who had invented the synthesis of ammonia, led the development of poison gas. Physicists worked on sound ranging to locate enemy guns and submarines.

The Second World War

Often called the physicists' war. Radar and the atomic bomb were developed by physicists in large, state-funded projects. At its peak the Manhattan Project employed more than 100,000 people. See The physics of nuclear weapons.

Big science

The term became popular after 1961, when Alvin Weinberg of Oak Ridge wrote about large-scale science requiring accelerators, reactors and rockets. Particle physics became the prototype: CERN was founded in 1954 partly to rebuild European physics and keep scientists in Europe, and Niels Bohr played a central role in its first years. In Denmark the research establishment at Risø was inaugurated in 1958.

The Cold War

When the Soviet Union launched Sputnik in 1957, the USA responded with NASA and a massive investment in science education. Throughout the Western world science teaching received a boost. Physics had become a matter of national security.

Big science also changed how research is done. The 2015 paper in which the ATLAS and CMS experiments at CERN jointly determined the mass of the Higgs boson had more than 5000 authors. Who is "the discoverer" when thousands have contributed? The Nobel Prize can be shared by at most three people.

Who was allowed to take part?

For most of history universities and academies were closed to women, and many other groups were kept out as well. Even so, women have contributed to physics – often without title, salary or recognition.

- **Émilie du Châtelet** (1706–1749) translated Newton's *Principia* into French with a commentary explaining it with Leibniz's calculus. Her translation is still standard in France.
- **Caroline Herschel** (1750–1848) discovered eight comets and in 1787 became the first woman to be paid for scientific work – 50 pounds a year from King George III as assistant to her brother William.
- **Marie Curie** (1867–1934) received the Nobel Prize in Physics in 1903 and in Chemistry in 1911. In 1911 the French Academy of Sciences refused to admit her.
- **Emmy Noether** (1882–1935) proved in 1918 the theorem that every symmetry in nature corresponds to a conservation law – one of the deepest results of theoretical physics. For several years she taught at Göttingen without pay and under Hilbert's name.
- **Lise Meitner** (1878–1968), together with her nephew Otto Frisch, explained in 1939 that the uranium nucleus splits. The Nobel Prize in Chemistry for the discovery went to Otto Hahn alone.
- **Chien-Shiung Wu** (1912–1997) showed by experiment in 1957 that nature distinguishes between right and left in the weak interaction. Lee and Yang, who had proposed it, received the Nobel Prize the same year.
- **Jocelyn Bell Burnell** (born 1943) discovered the first pulsars as a PhD student in 1967. The 1974 Nobel Prize went to her supervisor Antony Hewish and Martin Ryle.

Of the more than 220 recipients of the Nobel Prize in Physics, only five are women: Curie (1903), Maria Goeppert Mayer (1963), Donna Strickland (2018), Andrea Ghez (2020) and Anne L'Huillier (2023).

Politics could also close doors. After the Nazis seized power in 1933, Jewish scientists were dismissed from the German universities. Two Nobel laureates, Philipp Lenard and Johannes Stark, led a campaign for a so-called "German physics" against "Jewish" relativity. Many of the best fled, often with the help of networks such as the Danish Committee for the Support of Refugee Intellectual Workers, in which Harald and Niels Bohr were active. The loss to German science and the gain to the USA and Britain were enormous.

The norms of science – and their violation

In 1942 Robert K. Merton described four norms that he believed characterise science:

Communism (communality)

Results belong to everyone and must be published. In return the scientist receives recognition.

Universalism

A claim is judged on the evidence, not on who makes it – regardless of nationality, religion or sex.

Disinterestedness

Scientists should work for the truth, not for their own gain – and the system is arranged so that cheating seldom pays.

Organised scepticism

All results must be open to testing and criticism, including the established ones.

The norms are ideals, and history is full of breaches of them – from Newton's manipulated priority dispute with Leibniz to Jan Hendrik Schön's falsified results on organic superconductors, exposed in 2002. Merton also described the *Matthew effect* (1968): those who are already famous get more credit for the same work than unknowns – "unto every one that hath shall be given".

Physics, on the other hand, has led the way in making science open. In 1991 Paul Ginsparg started arXiv at Los Alamos, where physicists post their papers freely before they are refereed. Today this is the norm in large parts of physics, mathematics and computer science.

What is scientific knowledge?

If science is a social activity, is its knowledge also just a social construction? The question has divided philosophers, historians and scientists.

- **Karl Popper** (*Logik der Forschung*, 1934): science is recognised by the fact that its theories can be falsified. One can never prove a theory, only try to refute it.
- **Thomas Kuhn** (*The Structure of Scientific Revolutions*, 1962): most of the time scientists work within a paradigm – a shared set of theories, methods and exemplary problems – and do not try to falsify it. Only when anomalies pile up does a crisis and a revolution follow, as when quantum mechanics replaced classical physics.
- **Imre Lakatos**: theories are assessed as research programmes that are either progressive (predicting new things) or degenerating (saving themselves with auxiliary hypotheses). See the discussion of string theory.

- **Paul Feyerabend** (*Against Method*, 1975): there is no fixed scientific method; "anything goes", if one looks at what scientists have actually done.
- **The strong programme** (David Bloor, 1976): sociology must explain both true and false beliefs by the same kinds of cause.

In the 1990s the disagreement culminated in the so-called "science wars". In 1996 the physicist Alan Sokal got the cultural-studies journal *Social Text* to publish a deliberately meaningless article on the "hermeneutics" of quantum gravity. The hoax sparked a wide debate about how far one can go in describing natural science as a social construction.

A position on which many historians and physicists can agree today is that the *agenda*, institutions and language of science are shaped by society, while nature, through experiment and measurement, sets limits on which theories can survive. See also The scientific method does not exist.

A much-debated example is the historian Paul Forman's thesis of 1971: that the hostile intellectual climate of the Weimar Republic after the First World War – with its distrust of mechanical causal explanation – made German physicists ready to abandon determinism before quantum mechanics required it. The thesis is still disputed, but it shows how hard it is to draw a sharp line between the content of physics and its time.

Science and faith

In the 19th century John William Draper (1874) and Andrew Dickson White (1896) wrote influential books about an eternal war between science and religion. This so-called conflict narrative still shapes public debate. More recent historians of science, for instance in the anthology *Galileo Goes to Jail and Other Myths about Science and Religion* (ed. Ronald Numbers, 2009), have shown that the relationship was more complex. Many medieval natural philosophers were priests and monks, such as Albertus Magnus and Roger Bacon. The Jesuits ran observatories all over the world, and Clavius reformed the calendar. The Belgian priest Georges Lemaître proposed in 1927–31 the theory that became the Big Bang.

There have also been real conflicts, most famously the Galileo affair, but these were often as much about authority, politics and personalities as about faith. Many of the great names of physics, from Kepler and Newton to Faraday and Maxwell, were deeply religious, while others, such as Laplace and later Dirac, were not. How natural science and faith relate to each other is still a subject on which wise people disagree. See also The category error of the new atheism.

The responsibility of physicists

The atomic bomb made it impossible for physicists to regard themselves as neutral. During the war Niels Bohr tried to persuade Roosevelt and Churchill to share the secret with the Soviet Union to avoid an arms race, and in 1950 he sent an open letter to the United Nations about an

“open world”. In 1955 Albert Einstein and ten other scientists signed the Russell–Einstein Manifesto against nuclear war, which led to the Pugwash Conferences; in 1995 these received the Nobel Peace Prize together with the physicist Joseph Rotblat – the only scientist to leave the Manhattan Project on grounds of conscience.

Today the questions are different, but no less important. Climate physics – for which Syukuro Manabe and Klaus Hasselmann received the Nobel Prize in 2021 – has made physicists advisers on political decisions. In 2024 the Nobel Prize in Physics went to John Hopfield and Geoffrey Hinton for the physical ideas behind artificial neural networks, and Hinton used his speech to warn of the risks of artificial intelligence. Physics has never been remote from society; only the story told about it has sometimes been so.

Timeline: physics and society

1576 Tycho Brahe is given Hven by Frederick II.

1610 Galileo publishes *Sidereus Nuncius* and becomes court mathematician to the Medici.

1633 Galileo is condemned by the Inquisition.

1660 The Royal Society is founded in London.

1666 The Académie des sciences is founded in Paris with salaried members.

1675 The Royal Greenwich Observatory is founded to solve the longitude problem.

1687 Newton’s *Principia*.

1712 Newcomen’s steam engine pumps water out of coal mines.

1714 The Longitude Act: the state offers a large prize for solving a technical problem.

1759 Émilie du Châtelet’s translation of the *Principia* is published.

1787 Caroline Herschel becomes the first woman paid for scientific work.

1794 The École polytechnique is founded by the French Revolution to train engineers.

1824 Carnot asks how much work heat can give.

1829 The Polytechnic in Copenhagen is founded with H.C. Ørsted as its head.

1866 The first durable transatlantic telegraph cable.

1887 The Physikalisch-Technische Reichsanstalt is founded in Berlin.

1900 Planck introduces the energy quantum after measurements at the Reichsanstalt.

1911 Marie Curie receives her second Nobel Prize but is rejected by the French Academy.

1915 Poison gas is used in the First World War.

1921 Niels Bohr's institute opens on Blegdamsvej.

1933 Jewish scientists are dismissed from the German universities.

1939 Meitner and Frisch explain fission.

1942–45 The Manhattan Project.

1942 Merton describes the norms of science.

1945 Hiroshima and Nagasaki. Vannevar Bush's *Science, the Endless Frontier*.

1950 Niels Bohr's open letter to the UN.

1954 CERN is founded.

1955 The Russell–Einstein Manifesto.

1957 Sputnik. The Pugwash Conferences begin.

1957 Chien-Shiung Wu shows that parity is violated.

1958 The Risø research establishment is inaugurated.

1962 Kuhn's *The Structure of Scientific Revolutions*.

1967 Jocelyn Bell Burnell discovers pulsars.

1991 arXiv is started at Los Alamos.

1996 The Sokal affair.

2012 The Higgs boson is found at CERN by thousands of scientists.

2021 The Nobel Prize goes to climate physics.

2024 The Nobel Prize in Physics for the foundations of neural networks.

For teaching

The page can be used in physics lessons, in interdisciplinary courses and in project work, for example together with history or social studies. Some possible questions:

1. Choose a discovery from the biographies. Who paid for the work, and how did that affect what was investigated?
2. Hessen thought Newton's physics was a response to society's technical needs; Koyré, that it was a revolution in thought. What arguments can you find for each view?
3. Planck found his radiation law at an institute meant to help the lighting industry. Does that matter for whether the law is true?

4. The Nobel Prize can be shared by at most three people. Is that a reasonable rule in an age of papers with 5000 authors? How ought Jocelyn Bell Burnell's contribution to have been recognised?
5. Do physicists have a special responsibility for how their discoveries are used? Compare the atomic bomb with artificial intelligence.
6. Is the conflict narrative of science and religion a good description of history? Use the Galileo affair and Lemaître as examples.

Further reading

- Steven Shapin: *The Scientific Revolution* (Chicago 1996)
- Steven Shapin and Simon Schaffer: *Leviathan and the Air-Pump* (Princeton 1985)
- Thomas S. Kuhn: *The Structure of Scientific Revolutions* (Chicago 1962)
- Helge Kragh: *Quantum Generations. A History of Physics in the Twentieth Century* (Princeton 1999)
- Helge Kragh et al.: *Dansk Naturvidenskabs Historie* 1–4 (Aarhus University Press 2005–06, in Danish)
- Ronald L. Numbers (ed.): *Galileo Goes to Jail and Other Myths about Science and Religion* (Harvard 2009)
- Peter Galison and Bruce Hevly (eds.): *Big Science. The Growth of Large-Scale Research* (Stanford 1992)
- Margaret W. Rossiter: "The Matthew Matilda Effect in Science", *Social Studies of Science* 23 (1993)
- Hans Siggaard Jensen, Ole Knudsen and Frederik Stjernfelt (eds.): *Tankens magt – Vestens idéhistorie* (2006, in Danish)

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